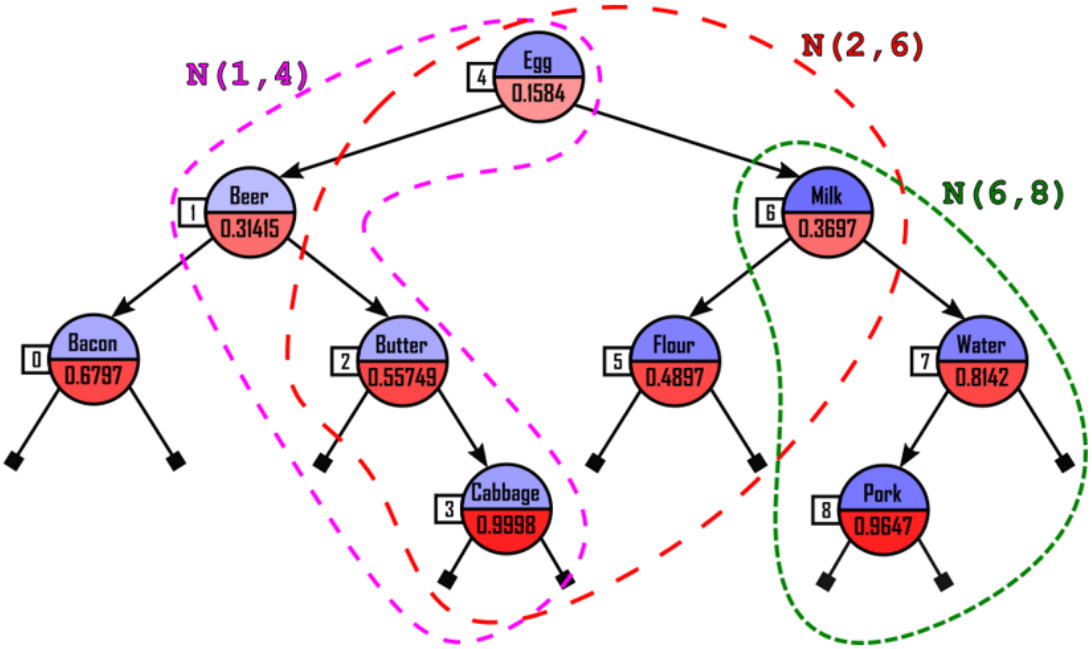


Figure 3.16 A few examples of the subsets of nodes `N(i,k)` defined in this section.

To compute the probability P(`AKi`), we need to introduce a new variable, and a lemma (an intermediate result).

We define `N(i,k) = N(k,i) = {Ni, Ni+1, … Nk-1, Nk}` as the subset of treap nodes whose keys are between the `i`-th and `k`-th smallest[11] of the whole tree.

Obviously, `N(0,n-1) = N(n-1,0)` contains all the nodes in the treap; figure 3.16 shows a few examples of these subsets, to help you visualize them: notice that the successor and predecessor of any node `N` always are on a path between N and the root, or `N` and a leaf; in other words, to find the predecessor – or successor – of a node, you only need to look either in the subtree rooted at `N` (where this node will be the left-most – or right-most, for successors), or in the path between `N` and the root.

It can be proved that the following lemma holds: